

ENSF 380

TOPIC 25: DATABASES PART I

Databases in Software Development

Traditional applications are numeric and textual databases

Databases are now used across all industries, such as:

- Multimedia databases
- Geographic Information Systems (GIS)
- Biological and genome databases
- Data warehouses
- Mobile databases
- Real-time and active databases
- Service industries
- Educational content and delivery

Social networks capture a lot of information about people and their interactions among people

- Posts, tweets, photos, videos in systems such as Facebook

Search engines (Google, Bing, Yahoo) collect their own repository of web pages for searching purposes

Basic Definitions

Data

- Known, recorded facts with an implicit meaning

Database

- A collection of related data

Mini-world

- Aspects of the real world described through data in a database
- e.g. student grades and transcripts at a university

Database Management System (DBMS)

- Software package/system that facilitates creation and maintenance of a computerized database

Database System

- DBMS software plus the data itself
- Applications may also be included

Application Activities

Applications interact with a database by generating queries and transactions

Queries access different parts of data and formulate the result of a request

Transactions may read some data and “update” certain values or generate new data and store that in the database

Applications must not allow unauthorized users to access data

Applications must keep up with changing user requirements against the database

SQL Overview

“Structured Query Language”

Language used to manage and retrieve data in relational databases

Relational databases represent data using tables (relations), rows (tuples), and columns (attributes)

Exercise 25.1

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Summary: Knowledge

Overview of relational databases

Basic terminology

Introduction to SQL

Familiarity with PostgreSQL command line and pgAdmin

Summary: Skills

Downloading and installing PostgreSQL

Basic queries and transactions with SQL commands

Creating a new user within PostgreSQL

Additional Information

Related reading

- Volume 2, Chapter 5.2: The Structured Query Language